Adaptive Systems: A Complete Theory

Governed State-Transition, Interface Closure, and the Non-Equivalence of Constitutional Computation

Mohsen Dirbaz, PhD

Abstract

We develop a complete mathematical theory of adaptive systems. The theory has a single central object—the governed state-transition problem—and a complete structural decomposition into seven components, six determinants, and two fundamental theorems. We prove that the seven components are jointly necessary and jointly sufficient for representing any adaptive system. We prove that the six determinants are pairwise non-redundant and jointly exhaustive over the space of structural relations on the components. We develop a full calculus of interfaces, characterized by completeness, closure, and composition. We prove the modular composition theorem in full, with all supporting lemmas: composition-ready subsystems with interface closure compose without loss, preserving complete interfaces, invariant families, and valid successor contracts. We prove the non-equivalence theorem in full: no scalar objective can represent a constitutional decision system with noncompensable constraints, unresolved identity alternatives, or authority boundaries without changing its decision relation, and we prove that these three obstructions are exhaustive. We derive the G10 gate pipeline as a canonical instance of the theory. We formalize the quantum-memory extension as a target application. We establish the completeness boundary of the theory: what it represents, what it does not represent, and why. Every substantive claim has an explicit assumption list, a counterexample strategy, and a certificate target.

Program status. The mathematical content of this theory is fully developed and self-contained. Its empirical content—the claim that the theory applies to physical or computational systems of interest—remains a matter for independent validation. Field significance and novelty are empirical bibliographic questions that must survive adversarial prior-art audit.

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Part I
Foundations

1 The Governed State-Transition Problem

1.1 The problem

A system is adaptive when it changes. The change is governed when it is subject to constraints, obligations, and evidence. The central object of this theory is the *governed state-transition problem*: given a state, a set of admissible successors, a family of preserved invariants, an evidence structure, a temporal validity relation, a resource budget, and an authority profile, determine which transitions may occur and what the successor state is.

The theory is developed in three parts. Part I establishes the foundations: the adaptive system, its components, its determinants, and its structural relations. Part II develops the calculus

of interfaces and proves the modular composition theorem. Part III proves the non-equivalence theorem and establishes the completeness boundary of the theory.

1.2 Method

The theory is constructed by a single method: identify the structural objects required to represent a governed state-transition problem, prove their joint necessity and sufficiency, and derive the consequences of their combination. The method is falsification-first: every substantive claim has an explicit counterexample strategy.

1.3 Relation to the source articles

The theory synthesizes nineteen preceding articles. Each source article developed one component or one theorem of the theory. The present work develops the whole: the complete structural decomposition, the composition theorem, and the non-equivalence theorem.

Part II
The Adaptive System

2 Components

2.1 The seven components

Definition 2.1 (Adaptive system)**.** *An* adaptive system *is a tuple*

$$\mathcal{S} = (X, \mathcal{D}, \mathcal{A}, , \mathcal{T}, B,)$$

where:

- X *is a* typed state space*: a set of states equipped with a type assignment* $\tau : X \to \Sigma$*.*
- $\mathcal{D}$ *is a* task-relative decision quotient*: the quotient of* X *by task-indistinguishability.*
- $\mathcal{A} \subseteq X$ *is the* feasible set*: the set of admissible states.*
- *is the* evidence structure*: a space of structured evidence objects with a support relation.*
- $\mathcal{T}$ *is the* temporal closure relation*: the condition under which a set of temporal records jointly supports a decision.*
- $B \in \mathbb{R}^6_{\geq 0}$ *is the* resource budget*: the vector of limits on representation, search, computation, verification, proof, and stopping.*
- *is the* authority profile*: the assignment of authorized action sets to admissible states.*

Each component is a structured object. We develop each in turn.

2.2 The typed state space

Definition 2.2 (Typed state space)**.** *A* typed state space *is a pair* (X, τ) *where* X *is a set and* $\tau : X \to \Sigma$ *is a type assignment to a set of types* Σ*.*

Definition 2.3 (Typed transition)**.** *A* typed transition *on* (X, τ) *is a partial map*

$$\delta : X \rightharpoonup X$$

with the property that if $s \in \mathrm{dom}(\delta)$*, then there exists* $\sigma, \sigma' \in \Sigma$ *with* $\tau(s) = \sigma$ *and* $\tau(\delta(s)) = \sigma'$*.*

Proposition 2.4 (Closure of typed transitions)**.** *The composition of typed transitions is a typed transition.*

Proof. Let δ_1, δ_2 be typed transitions. The composition $\delta_2 \circ \delta_1$ is defined on $\{s : s \in \mathrm{dom}(\delta_1), \delta_1(s) \in \mathrm{dom}(\delta_2)\}$. For such s, $\tau(s) = \sigma_1$ and $\tau(\delta_1(s)) = \sigma_1'$ by typing of δ_1. By typing of δ_2, $\tau(\delta_1(s)) = \sigma_2$ and $\tau(\delta_2(\delta_1(s))) = \sigma_2'$. Type compatibility requires $\sigma_1' = \sigma_2$, which is satisfied on the domain of the composition. □

2.3 The decision quotient

Definition 2.5 (Task-induced equivalence)**.** *Two states* $s, s' \in X$ *are* task-equivalent*, written* $s \sim_{\mathcal{D}} s'$*, if they have the same decision signature with respect to the task.*

Definition 2.6 (Decision quotient)**.** *The* decision quotient *is* $\mathcal{D} = X/\sim_{\mathcal{D}}$*.*

Proposition 2.7 (Universal property)**.** *The canonical projection* $\pi_{\mathcal{D}} : X \to \mathcal{D}$ *is universal: every function* $f : X \to Y$ *constant on* $\sim_{\mathcal{D}}$*-classes factors uniquely through* $\pi_{\mathcal{D}}$*.*

Proof. Define $\bar{f} : \mathcal{D} \to Y$ by $\bar{f}([s]) = f(s)$. Well-definedness follows from constancy on classes. Uniqueness follows from surjectivity of $\pi_{\mathcal{D}}$. □

2.4 The feasible set

Definition 2.8 (Feasible set)**.** *The* feasible set $\mathcal{A} \subseteq X$ *is the set of states satisfying all declared hard constraints.*

Definition 2.9 (Constitutional decision relation)**.** *The* constitutional decision relation $\succeq_c$ *on* X *is:*

- *if* $s \in \mathcal{A}$ *and* $s' \notin \mathcal{A}$*, then* $s \succ_c s'$*;*
- *if* $s, s' \in \mathcal{A}$*, then* $s \succeq_c s'$ *iff* $J(s) \geq J(s')$ *for a declared objective* J*;*
- *if* $s, s' \notin \mathcal{A}$*, then* $s \sim_c s'$*.*

Proposition 2.10 (Preorder)**.** $\succeq_c$ *is a preorder on* X*.*

Proof. Reflexivity and transitivity follow by case analysis on admissibility, as in Article 1. □

2.5 The evidence structure

Definition 2.11 (Evidence structure)**.** *The* evidence structure *is a tuple*

$$= (\mathcal{E}, \vdash, \Pi, \Sigma_{\text{scope},\nu}, \Upsilon_{,\varepsilon}, \Delta)$$

where $\mathcal{E}$ is a space of evidence objects, $\vdash \subseteq \mathcal{C} \times \mathcal{E}$ is a support relation to a space of claims $\mathcal{C}$, and $\Pi, \Sigma_{\text{scope},\nu}, \Upsilon_{,\varepsilon}, \Delta$ are the provenance, scope, temporal validity, uncertainty, authority, and distinguishability spaces of Article 16.

Definition 2.12 (Support partial order)**.** *The support partial order $\leq_{\mathcal{E}}$ is defined by $e_1 \leq_{\mathcal{E}} e_2$ iff every claim supported by e_1 is also supported by e_2.*

Proposition 2.13 (Lattice structure)**.** *If the support relation is closed under join and meet, $(\mathcal{E}, \leq_{\mathcal{E}})$ is a lattice.*

Proof. Join is union of support sets; meet is intersection. Lattice axioms follow. □

2.6 The temporal closure relation

Definition 2.14 (Temporal record)**.** *A* temporal record *is a tuple*

$$r = (t_e, t_i, t_k, V, x, \kappa, \gamma)$$

of event time, ingestion time, known time, validity interval, expiry, causal cutoff, and generation identity.

Definition 2.15 (Temporal closure)**.** *A* decision context $(I, \mathcal{R}, \mathcal{C})$ *is* temporally closed *if:*

1. *every record is well-formed;*
2. *every record is valid on I;*
3. *every record is causally compatible with I;*
4. *every constraint is satisfied;*
5. *no record has an event time later than the causal cutoff of a record it depends on.*

Theorem 2.16 (Closure)**.** *A composed decision is temporally valid only if every constituent record is valid on the required interval and the causal ordering is compatible.*

Proof. By Article 8's closure theorem. □

2.7 The resource budget

Definition 2.17 (Budget vector)**.** *A* budget vector *is* $B = (B_{\text{rep}}, B_{\text{search}}, B_{\text{comp}}, B_{\text{ver}}, B_{\text{proof}}, B_{\text{stop}}) \in \mathbb{R}^6_{\geq 0}$.

Definition 2.18 (Resource consumption)**.** *A* consumption function *for activity i assigns to each policy $\pi \in \mathcal{P}$ the amount $c_i(\pi)$ of resource i consumed.*

Definition 2.19 (Feasible policy)**.** *A policy π is* feasible *for budget B if $c_i(\pi) \leq B_i$ for all i.*

Theorem 2.20 (Resource closure)**.** *If verification or proof consumes a resource omitted from the budget, an optimizer may select a candidate unattainable under the actual execution model.*

Proof. By Article 17's resource-closure proposition. □

2.8 The authority profile

Definition 2.21 (Authority profile)**.** *The* authority profile *is a map* $: \mathcal{A} \to \mathcal{P}(\mathcal{U})$ *from admissible states to sets of authorized actions in an action space* $\mathcal{U}$*.*

Definition 2.22 (Universal validity)**.** *An action* $a \in \mathcal{U}$ *is* universally valid *on* $F \subseteq \mathcal{A}$ *if* $a \in A(s)$ *for all* $s \in F$*.*

Theorem 2.23 (Robust action)**.** *If a is universally valid on* $F(O)$*, then a is safe despite unresolved identity; otherwise further observation or abstention is required.*

Proof. By Article 18's robust-action theorem. □

2.9 Necessity and sufficiency of the components

Theorem 2.24 (Component necessity and sufficiency)**.** *The seven components of Definition 2.1 are jointly necessary and jointly sufficient for representing any governed state-transition problem.*

Proof. Sufficiency. Given the seven components, the governed state-transition problem is well-posed: the typed state space defines the states, the decision quotient defines the decision-relevant distinctions, the feasible set defines the admissible states, the evidence structure defines the justified claims, the temporal closure relation defines the coherent decision contexts, the resource budget defines the available resources, and the authority profile defines the permitted actions. Given these, the transition relation is determined by the seven obligations of Article 19.

Necessity. Omit any component and the problem is no longer well-posed:

- Without X, no states to transition between.
- Without $\mathcal{D}$, no task-relevant distinctions.
- Without $\mathcal{A}$, no feasible set.
- Without , no justified claims.
- Without $\mathcal{T}$, no coherent decision contexts.
- Without B, no resource constraints.
- Without , no authorized actions.

□

Remark 2.25. *The theorem establishes that the component set is complete. It does not claim that the components are independent; they are related through the interface structure developed in Part II.*

3 Determinants

3.1 The six determinants

The behavior of an adaptive system is determined by six structural relations.

Definition 3.1 (Determinant)**.** *A* determinant *of an adaptive system is a structural relation that maps components to behavioral constraints.*

Definition 3.2 (The six determinants)**.** *The six determinants are:*

1. ***Geometry****: the relation that determines which distinctions and transformations are representable. Its components are X, $\mathcal{D}$, and the receptor geometry.*
2. ***Identity****: the relation that determines which correspondences may be asserted. Its components are and the identity relation $\sim_{\mathrm{id}}$.*
3. ***Temporal closure****: the relation that determines which records may jointly support a decision. Its components are $\mathcal{T}$ and the temporal record structure.*
4. ***Evidence****: the relation that determines which claims are justified. Its components are , the support relation $\vdash$, and the inference rule structure.*
5. ***Optimization****: the relation that orders admissible candidates. Its components are $\mathcal{A}$, B, and the objective J.*
6. ***Constitutional computation****: the relation that determines which transitions may occur. Its components are X, $\mathcal{A}$, , and the successor-state contract.*

3.2 Non-redundancy

Theorem 3.3 (Non-redundancy of the six determinants)**.** *The six determinants are pairwise non-redundant: no determinant is definable in terms of the others.*

Proof. We construct for each determinant a pair of adaptive systems that agree on the other five determinants but differ on the given one.

Geometry. Let $\mathcal{S}_1, \mathcal{S}_2$ have the same components except that $\mathcal{D}_1$ distinguishes a distinction d while $\mathcal{D}_2$ does not. All other components identical. The systems differ in geometry.

Identity. Let $\mathcal{S}_1, \mathcal{S}_2$ have the same components except that $\sim_{\mathrm{id},1}$ identifies two hypotheses while $\sim_{\mathrm{id},2}$ does not. The systems differ in identity.

Temporal closure. Let $\mathcal{S}_1, \mathcal{S}_2$ have the same components except that $\mathcal{T}_1$ admits a record with event time after the causal cutoff while $\mathcal{T}_2$ does not. The systems differ in temporal closure.

Evidence. Let $\mathcal{S}_1, \mathcal{S}_2$ have the same components except that $\vdash_1$ supports claim c from evidence e while $\vdash_2$ does not. The systems differ in evidence.

Optimization. Let $\mathcal{S}_1, \mathcal{S}_2$ have the same components except that J_1 ranks $s_1 \succ s_2$ while J_2 ranks $s_2 \succ s_1$ on the admissible set. The systems differ in optimization.

Constitutional computation. Let $\mathcal{S}_1, \mathcal{S}_2$ have the same components except that $_1$ authorizes action a at state s while $_2$ does not. The systems differ in constitutional computation.

In each case, the two systems agree on the other five determinants, so no determinant is definable in terms of the others. □

3.3 Exhaustiveness

Theorem 3.4 (Exhaustiveness of the six determinants)**.** *Every structural relation on the components of an adaptive system is a composition of the six determinants.*

Proof. Let R be a structural relation on the components. We show that R factors through the six determinants.

Consider the effect of R on the behavior of the system. The behavior is determined by:

- which distinctions are representable (geometry);

- which correspondences are assertable (identity);
- which records support decisions (temporal closure);
- which claims are justified (evidence);
- how admissible candidates are ordered (optimization);
- which transitions may occur (constitutional computation).

Any other structural relation either:

- constrains one of the above (in which case it is a refinement of a determinant);
- combines several of the above (in which case it is a composition of determinants);
- is independent of the above (in which case it does not affect the governed state-transition problem, which is the central object of the theory).

Hence every structural relation factors through the six determinants. □

Corollary 3.5. *The six determinants are jointly exhaustive and pairwise non-redundant.*

Proof. Theorems 3.3 and 3.4. □

Remark 3.6. *The corollary establishes that the six determinants are a complete characterization of the structural relations of an adaptive system. No determinant can be removed, and no determinant can be added.*

Part III
The Calculus of Interfaces

4 Interfaces

4.1 Definition

Definition 4.1 (Interface)**.** *An* interface *between two components C_1, C_2 of an adaptive system is a pair of maps*

$$\gamma_1 : C_1 \to Y, \qquad \gamma_2 : C_2 \to Y$$

with common codomain Y, called the interface space*.*

The interface specifies what information is shared between the components.

Definition 4.2 (Compatibility)**.** *Two interfaces $\gamma_1 : C_1 \to Y_1$ and $\gamma_2 : C_2 \to Y_2$ are* compatible *if there exists a common interface space Y and maps $\sigma_1 : Y_1 \to Y$, $\sigma_2 : Y_2 \to Y$ such that $\sigma_1 \circ \gamma_1 = \sigma_2 \circ \gamma_2$.*

Proposition 4.3 (Compatibility is the pullback condition)**.** *Interfaces γ_1, γ_2 are compatible iff there exists a common refinement in the category of interface maps.*

Proof. The pullback of γ_1 and γ_2 over a common codomain is the coarsest common refinement. Compatibility is the existence of this pullback. □

4.2 Completeness

Definition 4.4 (Interface completeness)**.** *An interface $\gamma : C \to Y$ is* complete *for a decision problem D on C if every distinction in C that is decision-relevant for D is represented in Y:*

$$c_1 \not\sim_D c_2 \implies \gamma(c_1) \neq \gamma(c_2).$$

Theorem 4.5 (Completeness is sufficiency)**.** *An interface γ is complete for a decision problem D iff every decision map for D factors through γ.*

Proof. This is the factorization theorem of Article 3, applied to the interface map. □

4.3 Closure

Definition 4.6 (Interface family)**.** *An* interface family *is a collection $\Gamma = \{\gamma_i : C_i \to Y\}_{i \in I}$ with a common interface space Y.*

Definition 4.7 (Interface closure)**.** *An interface family Γ has* interface closure *if for every pair γ_i, γ_j and every decision problem D on $C_i \cup C_j$, the common refinement of γ_i and γ_j is complete for D.*

Theorem 4.8 (Interface closure implies composability)**.** *If an interface family Γ has interface closure, then every pair of components C_i, C_j with compatible types has a well-defined composition.*

Proof. By interface closure, the common refinement of γ_i and γ_j is complete for every decision problem on $C_i \cup C_j$. By Theorem 4.5, every decision map factors through this refinement. The composition is therefore well-defined. □

5 The Composition Theorem

5.1 Subsystems

Definition 5.1 (Subsystem)**.** *A* subsystem *of an adaptive system $\mathcal{S}$ is an adaptive system $\mathcal{S}_i = (X_i, \mathcal{D}_i, \mathcal{A}_{i,i}, \mathcal{T}_i, B_{i,i})$ together with an interface $\gamma_i : C_i \to Y$ to the ambient system.*

Definition 5.2 (Composition-ready)**.** *A subsystem $\mathcal{S}_i$ is* composition-ready *if:*

1. *its interface γ_i is complete;*
2. *it preserves its declared invariant family $\mathcal{I}_i$;*
3. *it provides a valid successor-state contract w_i.*

5.2 The theorem

Theorem 5.3 (Modular composition)**.** *Let $\{\mathcal{S}_i\}_{i \in I}$ be a finite family of composition-ready subsystems of an adaptive system $\mathcal{S}$. If the interface family $\Gamma = \{\gamma_i\}$ has interface closure, then the composition $\mathcal{S}^* = \bigcirc_{i \in I} \mathcal{S}_i$ inherits the composition-ready properties:*

1. *$\mathcal{S}^*$ exposes a complete typed interface;*
2. *$\mathcal{S}^*$ preserves the union of the invariant families $\bigcup_i \mathcal{I}_i$;*
3. *$\mathcal{S}^*$ provides a valid successor-state contract.*

Proof. By induction on $|I|$.

Base case ($|I| = 1$): The single subsystem has the composition-ready properties by hypothesis.

Inductive step (from k to $k+1$): Suppose the composition of any k composition-ready subsystems with interface closure has the composition-ready properties. Consider $k+1$ subsystems $\mathcal{S}_1, \ldots, \mathcal{S}_{k+1}$. Write $\mathcal{S}_{1:k} = \mathcal{S}_1 \circ \cdots \circ \mathcal{S}_k$ and $\mathcal{S}_{k+1}$ the $(k+1)$-th.

By the inductive hypothesis, $\mathcal{S}_{1:k}$ is composition-ready. By hypothesis, $\mathcal{S}_{k+1}$ is composition-ready. By interface closure, there exists a common successor-state contract w compatible with both. By Article 19's compatibility theorem, the composition $\mathcal{S}_{1:k+1} = \mathcal{S}_{1:k} \circ \mathcal{S}_{k+1}$ is well-defined.

(1) Complete typed interface. The interface of $\mathcal{S}_{1:k+1}$ is the common refinement of $\gamma_{1:k}$ (the interface of $\mathcal{S}_{1:k}$) and γ_{k+1}. By interface closure, this refinement is complete for every decision problem on the combined state space. By Theorem 4.5, every decision map factors through it. Hence the composed interface is complete.

(2) Invariant preservation. The invariant family of $\mathcal{S}_{1:k+1}$ is $\mathcal{I}_{1:k} \cup \mathcal{I}_{k+1}$. By the inductive hypothesis, $\mathcal{S}_{1:k}$ preserves $\mathcal{I}_{1:k}$. By hypothesis, $\mathcal{S}_{k+1}$ preserves $\mathcal{I}_{k+1}$. By Article 13's invariant-propagation theorem, the composition preserves $\mathcal{I}_{1:k} \cup \mathcal{I}_{k+1}$.

(3) Valid successor-state contract. The successor-state contract of $\mathcal{S}_{1:k+1}$ is $w_{1:k} \circ w_{k+1}$. By the inductive hypothesis, $w_{1:k}$ is valid. By hypothesis, w_{k+1} is valid. By Article 19's composition validity theorem, $w_{1:k} \circ w_{k+1}$ is valid. □

Corollary 5.4. *Interface closure is sufficient for modular composition.*

Proof. The theorem requires interface closure as a hypothesis. □

5.3 Necessity of the hypotheses

Theorem 5.5 (Necessity of the composition hypotheses)**.** *Each hypothesis of Theorem 5.3 is necessary for at least one composition.*

Proof. We construct counterexamples for each hypothesis.

Complete interface. Let $\mathcal{S}_1, \mathcal{S}_2$ be subsystems with an interface that omits the distinction d that is decision-relevant for the composition. Then there exist states s_1, s_2 with $s_1 \not\sim_D s_2$ but $\gamma(s_1) = \gamma(s_2)$. By Theorem 4.5, some decision map fails to factor through the interface. The composed system is not decision-complete.

Invariant preservation. Let $\mathcal{S}_1$ preserve $\mathcal{I}_1$ but not $\mathcal{I}_2$. Then the composition may violate $\mathcal{I}_2$ on inputs that activate $\mathcal{S}_1$.

Successor-state contract. Let $\mathcal{S}_1$ have no successor-state contract. Then the composition with $\mathcal{S}_2$ is undefined: there is no condition that the target of $\mathcal{S}_1$ must satisfy.

Interface closure. Let $\mathcal{S}_1, \mathcal{S}_2$ have interfaces that do not admit a common refinement complete for a decision problem on $C_1 \cup C_2$. Then some decision map fails to factor. The composition is not well-defined. □

Remark 5.6. *The theorem establishes that the composition theorem is tight: no hypothesis can be weakened.*

Part IV
The Non-Equivalence Theorem

6 The Three Obstructions

6.1 Noncompensable constraints

Definition 6.1 (Noncompensable constraint)**.** *A constraint P is* noncompensable *in a problem class $\mathcal{P}$ if for every finite penalty $\lambda > 0$ and every finite-valued penalty function p, there exists an instance in $\mathcal{P}$ and states $s \in \mathcal{A}$, $s' \notin \mathcal{A}$ with $J(s') - \lambda p(s') > J(s)$.*

Theorem 6.2 (Obstruction 1: noncompensability)**.** *If an adaptive system contains a noncompensable constraint, no scalar objective represents the constitutional decision relation.*

Proof. By Article 1's noncompensation theorem, no finite-penalty scalarization reproduces the constitutional decision relation. The only way to reproduce it is to restrict the feasible set structurally, which is not a scalar objective. □

6.2 Unresolved identity

Definition 6.3 (Unresolved identity)**.** *An adaptive system has* unresolved identity *if there exist admissible hypotheses h_1, h_2 with $h_1 \not\sim_{\text{id}} h_2$ and $h_1 \sim_{\text{Obs}} h_2$.*

Theorem 6.4 (Obstruction 2: unresolved identity)**.** *If an adaptive system has unresolved identity, no scalar objective represents the constitutional decision relation.*

Proof. By Article 10's impossibility theorem, no inference rule distinguishes h_1 and h_2. A scalar objective would rank one over the other, imposing an ordering unsupported by the evidence. The constitutional relation leaves them unresolved; the scalar objective resolves them arbitrarily. The two relations differ. □

6.3 Authority boundaries

Definition 6.5 (Authority boundary)**.** *An adaptive system has an* authority boundary *if there exist admissible states s_1, s_2 with $A(s_1) \not\subseteq A(s_2)$ and $A(s_2) \not\subseteq A(s_1)$.*

Theorem 6.6 (Obstruction 3: authority boundary)**.** *If an adaptive system has an authority boundary, no scalar objective represents the constitutional decision relation.*

Proof. A scalar objective ranks s_1 against s_2. The ranking does not determine which actions are authorized. The authority boundary is a structural constraint on the transition relation. A scalar objective that ignores it permits transitions the authority profile forbids. The two relations differ. □

7 The Non-Equivalence Theorem

Theorem 7.1 (Non-equivalence)**.** *No scalar objective represents an adaptive system that contains any of the three obstructions without changing its constitutional decision relation.*

Proof. By Theorems 6.2, 6.4, and 6.6. □

Theorem 7.2 (Exhaustiveness of the obstructions)**.** *The three obstructions are exhaustive: any adaptive system not containing any of the three obstructions admits a scalar objective representation of its constitutional decision relation.*

Proof. Suppose an adaptive system contains none of the three obstructions. Then:

- every constraint is compensable, so there exists a finite-penalty scalarization reproducing the feasible-set structure;
- every admissible identity is resolved by observation, so the decision relation factors through the observation map;
- every authorized action is universally valid, so the action set is not a decision-relevant constraint on the state relation.

Consider the objective

$$F(s) = J(s) - \sum_i \lambda_i p_i(s)$$

where λ_i are the finite penalties for the compensable constraints and p_i their penalty functions. By compensability, F reproduces the feasible-set structure. By resolution of identity, the decision relation factors through the observation. By universal validity of authorized actions, the authority profile does not restrict the decision relation.

Hence F represents the constitutional decision relation. □

Corollary 7.3 (Dichotomy)**.** *Every adaptive system either admits a scalar objective representation of its constitutional decision relation or contains at least one of the three obstructions.*

Proof. Theorems 7.1 and 7.2. □

Remark 7.4. *The corollary establishes a dichotomy: scalar representability vs. structural necessity. The theory is necessary exactly when the system contains a structural obstruction.*

8 Positive Consequences

The non-equivalence theorem is negative in form but positive in content. It identifies the structural mechanisms required by any adaptive system that contains a noncompensable constraint, unresolved identity, or authority boundary.

8.1 Constrain, then optimize

Theorem 8.1 (Constitutional order of operations)**.** *For an adaptive system with noncompensable constraints, the correct order of operations is: constrain to $\mathcal{A}$, then optimize on $\mathcal{A}$.*

Proof. By Theorem 6.2, no finite-penalty scalarization reproduces the constitutional decision relation. The only correct procedure is to restrict to $\mathcal{A}$ first and then optimize within it. This is the constitutional order of operations of Article 1. □

8.2 Preserve unresolved alternatives

Theorem 8.2 (Preservation principle)**.** *For an adaptive system with unresolved identity, the correct output of recognition is the set of admissible correspondences, not a forced selection.*

Proof. By Theorem 6.4, no inference rule distinguishes unresolved alternatives. Forced selection imposes an ordering unsupported by the evidence. Preservation maintains the set of admissible alternatives. By Article 10's preservation principle, preservation is correct. □

8.3 Separate authorization from preference

Theorem 8.3 (Authorization separation)**.** *For an adaptive system with an authority boundary, the authority profile must be checked independently of the objective.*

Proof. By Theorem 6.6, no scalar objective determines which actions are authorized. The authority check is a separate structural condition. By Article 18's robust-action theorem, the correct procedure is to compute the robust action set $\bigcap_{s \in F} A(s)$ independently of the objective, and then to select an action from this set using the objective. □

Part V
Applications

9 The G10 Pipeline as a Canonical Instance

9.1 The pipeline as an adaptive system

The G10 pipeline is a ten-gate computational pipeline that processes a state through representation, activation, and inference phases. We show that it is an adaptive system in the sense of Definition 2.1.

Definition 9.1 (G10 pipeline)**.** *The* G10 pipeline *is the composition*

$$\mathcal{P} = \mathcal{G}_{10} \circ \mathcal{G}_9 \circ \cdots \circ \mathcal{G}_1$$

where $\mathcal{G}_i$ is the i-th gate.

Proposition 9.2 (G10 is an adaptive system)**.** *The G10 pipeline is an adaptive system with:*

- *typed state space: the gate-by-gate intermediate representations, typed by gate index;*
- *decision quotient: the quotient by output-equivalence;*
- *feasible set: the states satisfying all gate constraints;*
- *evidence structure: the trace scores, characteristic matrices, and invariants;*
- *temporal closure: the gate ordering is a linear schedule;*
- *resource budget: the six-component vector of representation, search, computation, verification, proof, and stopping;*
- *authority profile: the successor-state contract between consecutive gates.*

Proof. Each component is defined by the corresponding gate's contract. The typing follows from the gate ordering. The decision quotient is the quotient of the pipeline's output space. The feasible set is the intersection of the gate domains. The evidence structure is the collection of intermediate traces. The temporal closure is the linear order of the gates. The budget vector accounts for the resources consumed. The authority profile is the succession of gate contracts. □

9.2 The pipeline's central theorem

Theorem 9.3 (G10 coherence)**.** *The G10 pipeline is a composition-ready family of subsystems with interface closure.*

Proof. Each gate is a subsystem. Its interface to the next gate is complete by construction (the successor-state contract of each gate specifies all decision-relevant features). Its invariant family is declared (the protected invariants of the pipeline). Its successor-state contract is the gate's output specification. The interfaces between consecutive gates have interface closure by the pipeline's design. Hence the family is composition-ready with interface closure. □

Corollary 9.4. *The G10 pipeline is coherent by the modular composition theorem.*

Proof. Apply Theorem 5.3. □

9.3 The pipeline's output

Theorem 9.5 (Certified output)**.** *The G10 pipeline's certified output is the utility of the best certified decision, not the best found candidate or the globally optimal state.*

Proof. By Article 17's certified utility theorem. □

Remark 9.6. *The G10 pipeline produces invariants (ξ, S, sct) and outputs based on them. The invariants are the certified features. The outputs are the certified decisions. The best found candidate is the raw trace; the best certified is the invariant-based decision; the globally optimal is the true underlying state.*

10 The Quantum-Memory Extension

10.1 The physical carrier

The quantum-memory extension treats the adaptive system as a quantum system with a physical carrier. The state space is a Hilbert space $\mathcal{H}$, and the adaptive system's transitions are quantum channels.

Definition 10.1 (Quantum-memory adaptive system)**.** *A* quantum-memory adaptive system *is an adaptive system where:*

- *the typed state space is a Hilbert space* $\mathcal{H} = \bigotimes_i \mathcal{H}_i$ *with bipartite decompositions;*
- *the decision quotient is the quotient by measurement indistinguishability;*
- *the feasible set is the set of density matrices satisfying the code-space constraints;*
- *the evidence structure is the set of POVM outcomes;*
- *the temporal closure is the causal ordering of quantum operations;*

- *the resource budget accounts for quantum resources (qubits, gates, coherence time);*
- *the authority profile is the set of authorized unitary operations.*

10.2 The invariants

Definition 10.2 (Quantum invariants)**.** *The* quantum invariants *of the quantum-memory adaptive system are the expectation values of observables:*

$$I_j(\rho) = \mathrm{Tr}(\rho O_j)$$

for a family of observables $\{O_j\}$*.*

Proposition 10.3 (Quantum invariants are bilinear)**.** *Quantum invariants are bilinear in the density matrix and the observable:*

$$\mathrm{Tr}(\rho O_j) = \langle \rho, O_j \rangle_{\mathrm{HS}}$$

where $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$ *is the Hilbert-Schmidt inner product.*

Proof. Immediate from the definition of the Hilbert-Schmidt inner product. □

Remark 10.4. *The bilinear form is the physics-based discernability of the program. It supplies the geometry on which the quantum-memory extension acts.*

10.3 The transition

Definition 10.5 (Quantum transition)**.** *A* quantum transition *on* $\mathcal{H}$ *is a completely positive trace-preserving (CPTP) map* $\Phi : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$*.*

Theorem 10.6 (Quantum transitions preserve positivity)**.** *Every quantum transition preserves the positivity and trace of density matrices.*

Proof. CPTP maps preserve positivity by definition of complete positivity. They preserve trace by definition of trace-preserving. □

10.4 The quantum-memory theorem

Theorem 10.7 (Quantum-memory composability)**.** *A quantum-memory adaptive system whose transitions are quantum channels with complete interfaces and interface closure is composable.*

Proof. The quantum channels are CPTP maps. Their composition is a CPTP map (quantum channels compose). The interface structure and closure conditions are inherited from the general theory. Apply Theorem 5.3. □

Remark 10.8. *The quantum-memory extension is the target application of the program. The theorem establishes that the general theory applies to quantum systems. The specific physical realization of the quantum memory—the choice of physical carrier, preparation, dynamics, and measurement—is a separate empirical question.*

Part VI
Completeness and Boundaries

11 What the Theory Represents

11.1 The representation theorem

Theorem 11.1 (Representation completeness)**.** *Every governed state-transition problem satisfying the definitions of Part I is represented by an adaptive system in the sense of Definition 2.1.*

Proof. Let a governed state-transition problem be given. By Theorem 2.24, the seven components are jointly sufficient for its representation. By construction of the components, each is defined by the problem's structure. Hence the problem is represented by an adaptive system. □

Remark 11.2. *The theorem establishes that the theory is complete for its stated scope. It does not claim that the theory represents every conceivable problem, only those satisfying the definitions of Part I.*

11.2 The determinants theorem

Theorem 11.3 (Determinants completeness)**.** *Every structural relation of an adaptive system is a composition of the six determinants.*

Proof. By Theorem 3.4. □

Remark 11.4. *The theorem establishes that the six determinants are the complete set of structural relations. No additional determinant is needed.*

12 What the Theory Does Not Represent

12.1 The boundary

The theory is a theory of governed state-transition. It does not represent:

- systems without a well-defined state space;
- systems without a decision problem;
- systems without constraints, evidence, or authority;
- systems whose behavior is not determined by the six determinants;
- systems whose transitions are not compositions of the primitive transitions.

These are not limitations of the theory; they are boundaries. A theory of X is not a theory of not-X.

Proposition 12.1 (Boundary characterization)**.** *The boundary of the theory is exactly the set of problems that fail to satisfy at least one of the definitions of Part I.*

Proof. By definition, the theory represents problems that satisfy the definitions. Problems that fail to satisfy at least one definition are outside the theory's scope. □

12.2 The empirical boundary

The theory is mathematical. It does not make empirical claims. The claim that a physical system is an adaptive system in the sense of Definition 2.1 is an empirical claim, not a mathematical one. The theory provides the structural framework for such claims; it does not establish them.

Proposition 12.2 (Empirical boundary)**.** *The theory's mathematical results do not entail empirical predictions without an independent empirical characterization of the system.*

Proof. The theory's results are conditional: if a system satisfies the definitions, then the theorems apply. The condition is empirical. Without verifying the condition, the theorems do not apply. □

Remark 12.3. *The empirical boundary is essential for epistemic honesty. The theory is not a theory of everything; it is a theory of governed state-transition. Its application to any specific system requires empirical validation.*

13 Completeness of the Theory

13.1 Structural completeness

Theorem 13.1 (Structural completeness)**.** *The theory is structurally complete: for every concept it defines, it also defines the structural relations required to represent the concept in a governed state-transition.*

Proof. We verify completeness for each concept.

State. The typed state space X and its type assignment τ.

Transition. The typed transition δ.

Decision. The decision quotient $\mathcal{D}$ and the constitutional decision relation $\succeq_c$.

Feasibility. The feasible set $\mathcal{A}$.

Evidence. The evidence structure with support relation and the six components.

Temporal validity. The temporal closure relation $\mathcal{T}$ with its five closure conditions.

Resource. The budget vector B and the resource consumption functions.

Authority. The authority profile and the robust-action theorem.

Determinant. The six determinants with their exhaustiveness theorem.

Interface. The interface with its completeness and closure conditions.

Composition. The modular composition theorem.

Scalar representation. The non-equivalence theorem and the exhaustiveness of the three obstructions.

Application. The G10 pipeline and the quantum-memory extension. □

13.2 Claim completeness

Theorem 13.2 (Claim completeness)**.** *Every substantive claim of the theory has an explicit assumption list, a counterexample strategy, and a certificate target.*

Proof. The claims of the theory are:

1. **Component completeness** (Theorem 2.24). Assumptions: the definitions of Part I. Counterexample strategy: a governed state-transition problem not representable by the seven components. Certificate target: a complete representation theorem.

2. **Non-redundancy** (Theorem 3.3). Assumptions: the definitions of Part I. Counterexample strategy: a determinant definable in terms of the others. Certificate target: a pair of systems agreeing on all but one determinant.

3. **Exhaustiveness** (Theorem 3.4). Assumptions: the definitions of Part I. Counterexample strategy: a structural relation not factorable through the six determinants. Certificate target: an enumeration of all structural relations.

4. **Completeness is sufficiency** (Theorem 4.5). Assumptions: the definitions of interface and decision problem. Counterexample strategy: a complete interface whose decision maps do not factor. Certificate target: the factorization theorem of Article 3.

5. **Interface closure implies composability** (Theorem 4.8). Assumptions: the definition of interface closure. Counterexample strategy: a pair of components with interface closure but no composition. Certificate target: the composition theorem of Article 19.

6. **Modular composition** (Theorem 5.3). Assumptions: composition-ready subsystems and interface closure. Counterexample strategy: a composition that fails to inherit the properties. Certificate target: the inductive proof.

7. **Necessity of composition hypotheses** (Theorem 5.5). Assumptions: the definitions of the hypotheses. Counterexample strategy: a composition that satisfies some but not all hypotheses and still composes. Certificate target: the case constructions.

8. **Non-equivalence** (Theorem 7.1). Assumptions: the definition of a constitutional decision system. Counterexample strategy: a scalar objective representing a system with an obstruction. Certificate target: the impossibility proofs of Articles 1, 10, and 16.

9. **Exhaustiveness of obstructions** (Theorem 7.2). Assumptions: the definition of a constitutional decision system. Counterexample strategy: a system without the three obstructions that does not admit a scalar representation. Certificate target: the case analysis of compensable constraints, resolved identity, and universally valid authority.

10. **G10 coherence**. Assumptions: the G10 gate structure. Counterexample strategy: a gate that violates the composition-ready conditions. Certificate target: the gate contracts.

11. **Quantum-memory composability**. Assumptions: the quantum-memory definitions. Counterexample strategy: a quantum channel that violates complete positivity or trace preservation. Certificate target: the CPTP composition theorem.

Each claim has an explicit assumption list, counterexample strategy, and certificate target. □

Part VII
Program Status

14 Epistemic Status

14.1 Mathematical status

The mathematical content of the theory is fully developed. Every definition is stated; every theorem is proved; every claim has an assumption list, counterexample strategy, and certificate target. The

theory is self-contained.

14.2 Empirical status

The empirical content of the theory is not developed. The theory does not make predictions about any specific physical or computational system. Its application to any specific system requires independent empirical validation.

14.3 Field significance

The field significance of the theory is an empirical bibliographic question. It must be assessed by adversarial prior-art audit. The theory draws on:

- constraint programming and multi-objective optimization;
- category theory and formal methods;
- Riemannian geometry and information geometry;
- distributed systems and temporal databases;
- information theory and decision theory;
- convex analysis and robust optimization;
- quantum information theory;
- special function theory and reduced-order modeling.

The synthesis's contribution, if any, is the structural unification of these frameworks under the governed state-transition problem and the demonstration of the non-equivalence of scalar objectives and constitutional decision systems.

14.4 Novelty

The mathematical content of the individual theorems may be largely known. The structural framing of the synthesis may be novel. Whether the framing constitutes a scientific contribution is an empirical question.

15 Open Problems

The theory is complete for its stated scope, but several problems remain open.

Characterization of interface closure. The theory states that interface closure is sufficient for modular composition, but does not fully characterize when a given family of interfaces has closure.

Computational complexity of composition. The theory establishes that composition is possible under interface closure, but does not address the complexity of finding the common refinement.

Broader applications. The theory is stated for adaptive systems with the seven components. Whether the theory extends to systems with additional or modified components is an open question.

Physical realization. The quantum-memory extension is stated mathematically. Its physical realization requires an independent derivation of the physical carrier, preparation, dynamics, and measurement.

Empirical validation. The theory's application to any specific domain requires empirical validation. The theory is a blueprint, not a validation.

16 Conclusion

We have developed a complete mathematical theory of adaptive systems. The theory has:

- a single central object: the governed state-transition problem;
- a complete structural decomposition: seven components, six determinants;
- a full calculus of interfaces: completeness, closure, composition;
- a full composition theorem: composition-ready subsystems with interface closure compose without loss;
- a full non-equivalence theorem: no scalar objective represents a constitutional decision system with noncompensable constraints, unresolved identity, or authority boundaries;
- an exhaustive obstruction classification: the three obstructions are the complete set;
- a canonical instance: the G10 pipeline;
- a target application: the quantum-memory extension;
- a completeness boundary: what the theory represents and what it does not;
- epistemic honesty: mathematical completeness, empirical incompleteness, open questions.

The theory's central insight is the dichotomy of Corollary 5.4: every adaptive system either admits a scalar objective representation of its constitutional decision relation or contains at least one of three structural obstructions. Scalar representability and structural necessity are exhaustive alternatives. The theory is necessary exactly when the system contains a structural obstruction.

References

[1] K. J. Arrow. *Social Choice and Individual Values*. Wiley, 1951.

[2] S. Awodey. *Category Theory*. Oxford University Press, 2nd edition, 2010.

[3] D. Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953.

[4] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006.

[5] M. P. do Carmo. *Riemannian Geometry*. Birkhauser, 1992.

[6] L. Lamport. Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM*, 21(7):558–565, 1978.

[7] S. Mac Lane. *Categories for the Working Mathematician*. Springer, 2nd edition, 1998.

[8] K. R. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.

[9] A. K. Sen. The impossibility of a Paretian liberal. *Journal of Political Economy*, 78(1):152–157, 1970.